

Software Architecture & Design (SAD) Deliverables - Ritu

SAD 3.4 Component Description

Processing/Status Module:

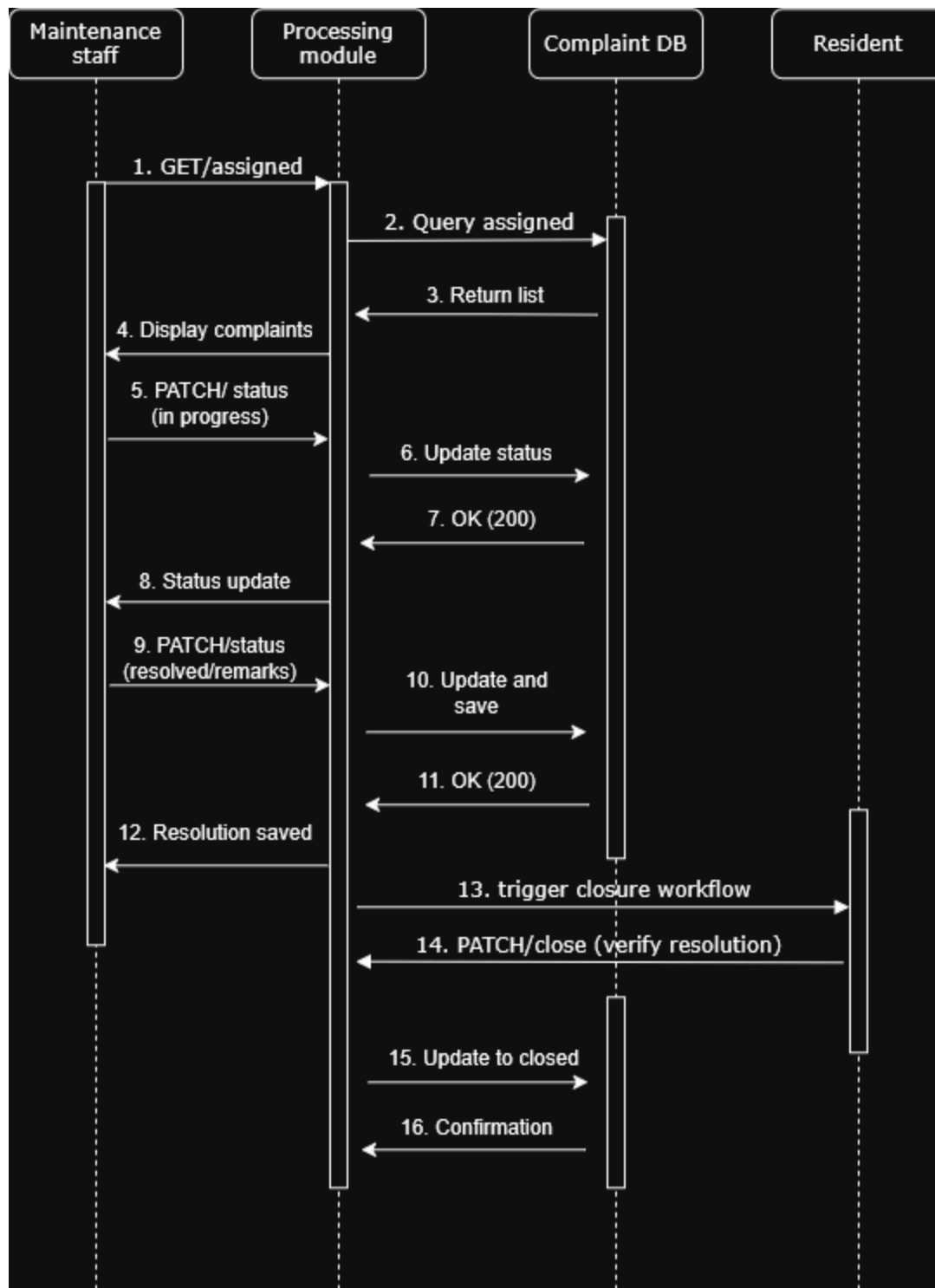
The Processing/Status Module manages the entire lifecycle of complaints after assignment, handling state transitions through the defined workflow sequence: Submitted -> Assigned -> In Progress -> Resolved -> Closed. It enables maintenance staff to view assigned tasks, update work statuses, attach diagnostic/resolution remarks, and record completed resolution details. Furthermore, it strictly enforces business rules for complaint closure, ensuring that a complaint cannot transition to Closed without documented resolution details and prohibiting direct closures by unauthorized actors.

SAD 3.8 Traceability Matrix Entries

Requirement ID	Component	Test Case IDs
FR-012	Processing/Status Module	TC-PRC-01, TC-PRC-02
FR-013	Processing/Status Module	TC-PRC-03, TC-PRC-04
FR-014	Processing/Status Module	TC-PRC-05, TC-PRC-06
FR-020	Processing/Status Module	TC-PRC-07, TC-PRC-08
NFR-001	Application Infrastructure	TC-PERF-01, TC-PERF-02
NFR-002	Processing/Status Module	TC-NFR-01
NFR-003	Processing/Status Module	TC-NFR-02

NFR-004	Processing/Status Module	TC-NFR-03
NFR-005	Application Architecture	TC-NFR-04
NFR-006	Application Architecture	TC-NFR-05

SAD 4.2 Sequence Diagram: Process → Resolve → Close Workflow



SAD 4.3 API Definitions for Processing/Status Module

1. View Assigned Complaints

Endpoint: /api/v1/staff/complaints/assigned

Method: GET

Request: {status}

Response: {status, data: [{complaint_id, title, category, location, status, created_at}]}

Errors: 401 Unauthorized, 403 Forbidden

2. Close Complaint Workflow

Endpoint: /api/v1/staff/complaints/{complaint_id}/status

Method: PATCH

Request: {status, remarks, resolution_details}

Response: {status, message, data: {complaint_id, status, updated_at}}

Errors: 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found

3. Update Complaint Status & Resolution Remarks

Endpoint: /api/v1/complaints/{complaint_id}/close

Method: PATCH

Request: {confirmation, closure_notes}

Response: {status, message, data: {complaint_id, status, closed_at}}

Errors: 400 Bad Request, 403 Forbidden

SAD 4.4 Error Handling, Logging & Monitoring

Standardized Error Handling:

All application errors produce:

error_code: INVALID_STATE_TRANSITION

message: Cannot transition complaint state from Submitted directly to Resolved.

timestamp: 2026-10-02T20:50:00Z

Logging Standards:

Structured logging via SLF4J/Logback in JSON format. Sensitive data, including passwords, session tokens, and PCI/PII data, is sanitized using logging filters prior to serialization.

Monitoring Metrics:

Application metrics are collected using Micrometer/Prometheus endpoints monitoring API latency (p95, p99 target < 10s under load), transition failure counts, and overall system availability metrics.